

LECTURE L9-10 • MODULE 2 • CO2

Bivariate relationships

Week 5 | Slides: DATA209_Module_2_Core_Advanced_EDA.pptx, pages 11-20 | Laboratory: P9-10

TOPICS COVERED

- Correlation and covariance
- Scatterplots and what they hide
- Relationships between categorical variables

Covariance and correlation

Covariance is the average product of paired deviations from the two means. Its sign gives the direction; its magnitude is uninterpretable because it carries the units of both variables. Changing metres to centimetres changes the number without changing the relationship.

Correlation is covariance divided by the two standard deviations: bounded in $[-1, 1]$, unitless, and therefore comparable across pairs. That comparability is the entire point.

Choosing a coefficient

COEFFICIENT	MEASURES	USE WHEN	ROBUST
Pearson r	linear association	both continuous, roughly linear	no
Spearman ρ	monotonic association on ranks	non-linear but consistently increasing; ordinal	yes
Kendall τ	concordance of pairs	small samples; many ties	yes
Cramér's V	association between categoricals	two nominal variables	n/a
Correlation ratio η	categorical vs numeric	group means differ across a category	partly

Comparing Pearson with Spearman is a cheap non-linearity test: a large gap means the relationship is monotonic but not straight.

Correlation is not causation, and not shape

Correlation measures co-movement. It says nothing about which variable moved first, or whether a third moved both. And it says nothing about shape — hence the rule:

A correlation coefficient reported without its scatterplot is an unverified claim.

Five patterns worth naming when you look: **linear**, **non-linear** (Pearson understates it), **heteroscedastic** (spread widens along the axis), **clustered** (a single correlation over all points is meaningless — split first), and **outlier-driven** (a few points create or destroy the association; recompute without them and compare).

Relationships involving categories

- **Numeric vs categorical** — grouped boxplots or violins; compare medians *and* spreads, not just means. Quantify with a group-means table.

- **Categorical vs categorical** — cross-tabulation with **row or column percentages**, never raw counts. Quantify with chi-square, then Cramér's V for strength.

Percentages beat counts: 20 purchases from 40 sessions beats 100 from 1,000, and raw counts hide that. Always report the denominator alongside the rate.

With 12,330 rows almost every chi-square test returns a significant p-value. Read the **effect size**, not the p-value.

From pattern to hypothesis

A hypothesis names a **mechanism**, predicts a **direction**, and states what would **falsify** it.

- Weak: "Glucose is related to the outcome." This restates the correlation and predicts nothing.
- Better: "Elevated fasting glucose precedes diagnosis, so glucose will separate the classes better

than BMI — falsified if the separation vanishes after conditioning on age."

Name the confounders you cannot rule out with this data. Age confounds almost everything in the Pima dataset; the promotional calendar is unobserved in the shoppers dataset.

KEY TERMS

Covariance	Pearson correlation	Spearman correlation	Kendall tau	Cramér's V
Heteroscedasticity	Cross-tabulation	Chi-square test	Effect size	Confounder

COMMON ERRORS TO AVOID

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- Running Pearson correlation on a nominal or ordinal column
 - Reporting a correlation without plotting the pair
 - Comparing raw counts across groups of very different sizes
 - Treating a small p-value on 12,000 rows as evidence of a large effect

READINGS — ALL FREE TO ACCESS

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1. scikit-learn — clustering and metrics
<https://scikit-learn.org/stable/modules/clustering.html>
 2. seaborn — relational and categorical plots
<https://seaborn.pydata.org/tutorial.html>
 3. James et al. — Introduction to Statistical Learning
<https://www.statlearning.com/>

TEST YOURSELF

1. Pearson $r = 0.15$ and Spearman $\rho = 0.61$ for the same pair. What does that tell you?
2. Why is covariance rarely reported directly?
3. Two categories show conversion rates of 50% and 10%. What must you also report?
4. Rewrite 'sales correlate with advertising' as a falsifiable hypothesis.